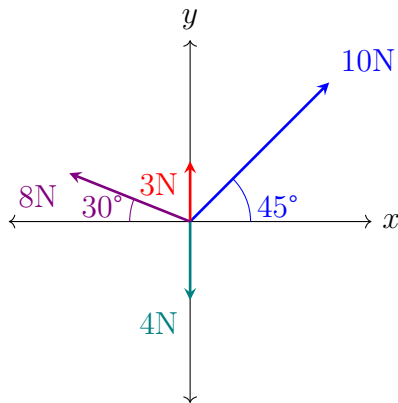


Mechanics Revision Questions

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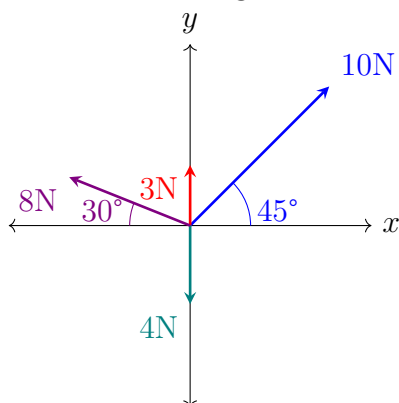
1. Find the magnitude and direction of $\mathbf{r} = -4\mathbf{i} + 7\mathbf{j} = \begin{pmatrix} -4 \\ 7 \end{pmatrix}$.
2. The vector $\mathbf{r}$ has length 360cm and angle 35° below the x-axis. Express $\mathbf{r}$ as $\begin{pmatrix} a \\ b \end{pmatrix}$ in S.I. units.
3. A particle is at the origin. It moves with constant velocity $\begin{pmatrix} 3 \\ 5 \end{pmatrix} \text{ms}^{-1}$ for 5 seconds, then with constant velocity $\begin{pmatrix} 2 \\ -7 \end{pmatrix} \text{ms}^{-1}$ for 2 seconds. Find its final position and distance from the origin.
4. One particle leaves the origin with velocity $(10\mathbf{i} + 7\mathbf{j})\text{ms}^{-1}$. Another leaves from the coordinates $(40, 7)$ with velocity $(5\mathbf{i} + 5\mathbf{j})\text{ms}^{-1}$. Find the time and position of collision.
5. Particle A has position vector $\mathbf{r}_A = \begin{pmatrix} 3t \\ 4t \end{pmatrix}$ and particle B has position vector $\mathbf{r}_B = \begin{pmatrix} 14 - t \\ 3 + t \end{pmatrix}$.
 - (a) Find $\overrightarrow{AB}$.
 - (b) Find the distance between A and B at time t .
 - (c) Find the minimum distance between A and B, and the time at which this occurs.
6. A particle passes through $X = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ with speed $\begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ms}^{-1}$ and acceleration $\begin{pmatrix} 8 \\ -4 \end{pmatrix} \text{ms}^{-1}$. When is the particle 3m from X ?
7. A particle begins travelling at 5ms^{-1} . After 3s it accelerates at 2ms^{-1} for 6s and then decelerates at 3ms^{-1} until it stops.
 - (a) Find the maximum speed of the particle.
 - (b) When does the particle stop?
 - (c) Sketch the speed-time graph for the particle.
 - (d) Find the distance travelled by the particle.

8. There are two forces $\mathbf{F}_1$ and $\mathbf{F}_2$ are both 10N in magnitude. The angle between them is 70° . Find the resultant force.
9. Consider the diagram drawn below.



- (a) Find the resultant force.
- (b) An additional force $\mathbf{P}$ is applied with angle 180° above the x-axis, and another force $\mathbf{Q}$ is applied with angle 60° above the x-axis. If the resultant force is zero, then what are $\mathbf{P}$ and $\mathbf{Q}$?
10. A particle of mass 16kg rests in equilibrium on a plane at an angle of 45° to the horizontal. Find the magnitude of the friction and the normal reaction.
11. A particle of mass 5kg lies in equilibrium on a slope of angle 20° . The coefficient of friction is 0.4 (take $g=9.8\text{kgN}^{-1}$).
- (a) Find the magnitude of the friction force.
- (b) The angle of the slope is gradually increased. At what angle will the particle begin to roll?
12. A box of 50kg is being pulled along the ground by a horizontal force of 165N. If the acceleration is 2ms^{-1} , then find the coefficient of friction.
13. A particle of mass 19000g is being pulled up a 30° slope by a rope at 15° to the slope. The tension of the rope is 175N and the coefficient of friction is 0.3. Find the particle's acceleration.
14. A rescue vehicle of mass 2000kg is towing a broken down car of mass 950kg. The rescue vehicle has 1750N driving force. Assuming no resistance calculate the following.
- (a) The acceleration of the combined vehicles.
- (b) The force in the tow bar.
15. Two particles of masses 10kg and 4kg respectively are attached to a light and inextensible string which lies over a smooth pulley. The 10kg particle is 95cm above the ground and 300cm above the pulley. The 4kg particle is 50cm above the ground. When both particles are released what is the maximum height achieved by each of the particles.

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